

Trends in Conservative Vote Choice

1. Background

Political scientists have long used logistic regression to study how demographic characteristics predict vote choice, and how those relationships evolve over time. A classic version of this exercise uses the U.S. National Election Study (NES) to model the probability of voting Republican from 1952 to 2000, refitting the same model separately for every four-year election cycle and tracking how each predictor's coefficient moves over time.

This report adapts that design to a British context using the British Election Study (BES) Internet Panel, a large repeated-measures survey that re-interviews the same respondents across 26 waves. Rather than a four-yearly election survey, the BES panel took snapshots at irregular intervals from roughly 2014 to 2021 — a period spanning the 2015, 2017 and 2019 general elections and the 2016 EU referendum.

2. Dataset Description

Variable	Category	N
Age	18-49	11.011
	50-64	15.704
	65+	19.592
Education	1- No qualifications / below GCSE	5.921
	2- GCSE	10.264
	3- A-level	8.368
	4- Undergraduate/Postgraduate	21.754
Ethnicity	0- White	44.759
	0.5- Other	1.351
	1- Black	197
Sex	Male	26.406
	Female	19.901

3. Method

For each of the 26 BES panel waves, a separate logistic regression was fit on that wave's respondents:

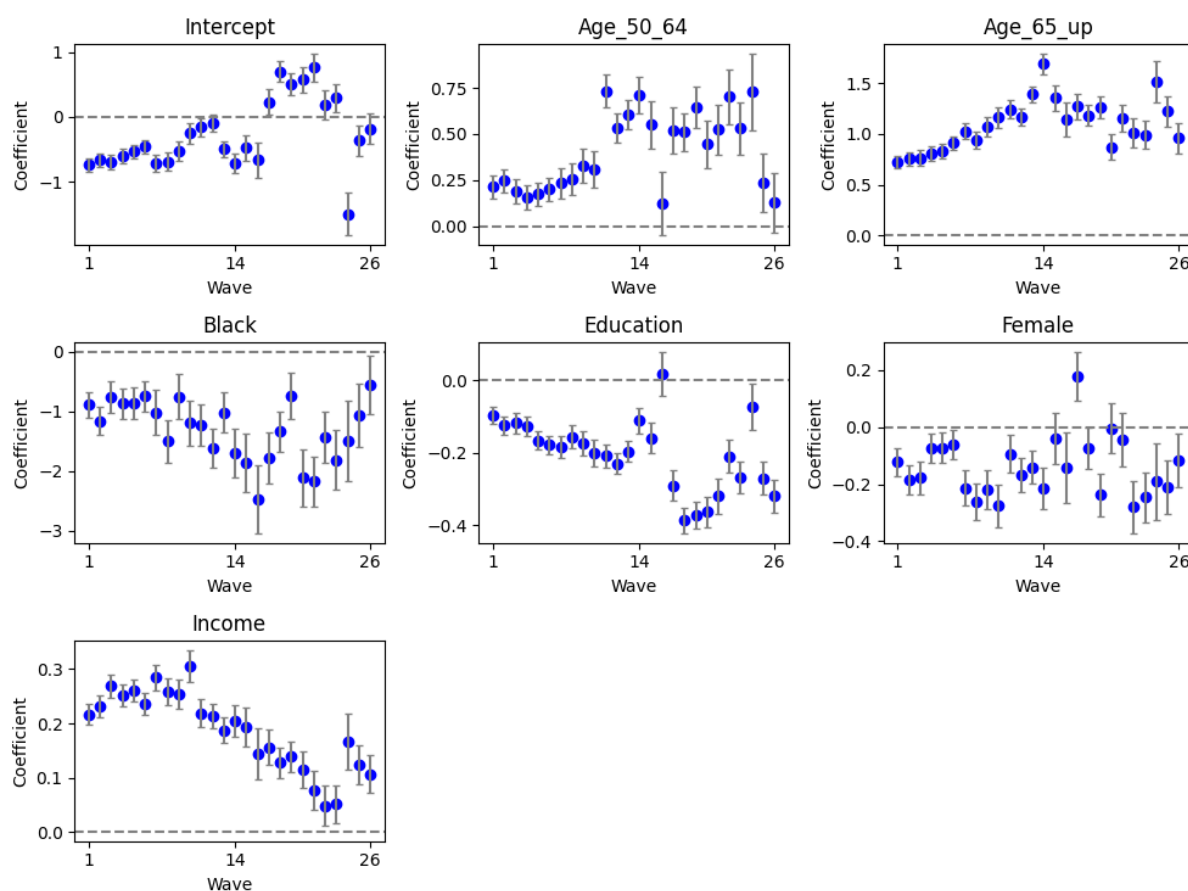
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convote ~ C(age_discrete) + ethnicity_adj + educ1 + female + income
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Age is entered as a categorical variable with the 18–49 group as the reference category, so its two coefficients represent the change in log-odds of voting Conservative relative to that youngest group. All 26 wave-level models converged without numerical instability

once the age variable was collapsed to three categories. The average pseudo R² across waves was 0.055, typical for individual-level vote-choice models built from a small set of demographic predictors — demographics explain part, but far from all, of the variation in vote choice.

4. Results

The chart below shows the estimated coefficient for each predictor in every wave, plotted with error bars of ± 0.67 standard errors (an approximate 50% confidence interval). Points above the dashed zero line indicate the predictor is associated with a higher chance of voting Conservative; points below indicate a lower chance.



5. Interpretation

- Income is one of the most consistent predictors across the whole panel: higher household income quintile is associated with a higher probability of voting Conservative in every single wave. The effect is strongest in the earlier waves (around 2014–2016) and gradually weakens through the later waves, though it never crosses zero.
- Age shows a clear, stable gradient: the 50–64 and, especially, the 65+ group are consistently more likely to vote Conservative than the 18–49 reference group, and

the size of this gap is one of the largest effects in the model throughout the panel — echoing the age effect found in the original American analysis.

- Ethnicity has a large, consistently negative coefficient: Black respondents are markedly less likely to vote Conservative than White respondents across every wave. The confidence intervals are wide because Black respondents make up under half a percent of this sample, so this estimate should be read as directionally robust but imprecise.
- Education is small but consistently negative, and this effect visibly strengthens (becomes more negative) from around the middle waves onward — consistent with education becoming a sharper political dividing line in British politics around and after the 2016 EU referendum.
- Sex shows a small, mostly negative but noisy coefficient — women are somewhat less likely than men to vote Conservative in most waves, but the effect is inconsistent and often not distinguishable from zero, echoing the fluctuating gender effect in the original NES case.
- The intercept itself drifts substantially across waves, reflecting swings in overall Conservative support that are common to all groups — dipping sharply in the final two or three waves, consistent with the fall in the raw Conservative vote share seen in those waves (41–46%, down from a panel-wide average of 53%).